Huawei Access Node JAY-ACC-35 Routing Protocol Flap SOP

Node: JAY-ACC-35 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_9910

Description

The Huawei Access Node JAY-ACC-35 is configured to operate as a core routing device running BGP, OSPF, and ISIS protocols. This architecture ensures high availability and efficient routing of traffic between various network segments. However, BGP route churn, OSPF adjacency loss, and ISIS LSP flooding can occur due to misconfigurations, hardware failures, or software issues. In this SOP, we will troubleshoot and resolve the critical alarm ALARM_9910, which indicates a Routing Protocol Flap in the Access Node.

Pre-requisite Checks

- Verify that the site is accessible and the remote hands are dispatched to Jayanagar Metro Site.

- Check the redundant link status to ensure that it is operational.

- Confirm that the Huawei CLI access is established and the user is logged in with the necessary privileges.

Potential Root Causes

Cause	Probability
BGP Configuration Error	High
OSPF Neighborship Issues	Medium
ISIS LSP Flooding Due to Hardware Failure	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp all summary, Expected Output: BGP Summary: Total peer: 10, Total AS: 100, Description: This command displays a summary of BGP peerings and AS numbers. It helps to identify any errors or inconsistencies in BGP configuration.

Command: display ospf neighbor, Expected Output: OSPF Neighbor: 5.5.5.5, State: FULL, Description: This command displays the OSPF neighbor table. It helps to identify any OSPF adjacency issues or errors.

Command: display isis database, Expected Output: ISIS Database: 10 LSPs, 5 Areas, Description: This command displays the ISIS database. It helps to identify any ISIS LSP flooding or errors.
```

Resolution Steps

Step 1: Step: Re-Configure BGP Peerings, Command: reset bgp peer 10.10.10.10, Description: This step re-configures BGP peerings to resolve BGP configuration errors.

Step 2: Step: Reset OSPF Neighborship, Command: reset ospf neighbor 5.5.5.5, Description: This step resets OSPF neighborship to resolve OSPF adjacency issues.

Step 3: Step: Clear ISIS LSP Flooding, Command: clear isis database, Description: This step clears ISIS LSP flooding due to hardware failure.

Post-Check Verification

```
Command: display bgp all summary, Expected Output: BGP Summary: Total peer: 10, Total AS: 100, Description: This command verifies that BGP peerings and AS numbers are correct after re-configuration.

Command: display ospf neighbor, Expected Output: OSPF Neighbor: 5.5.5.5, State: FULL, Description: This command verifies that OSPF neighborship is correct after re-configuration.

Command: display isis database, Expected Output: ISIS Database: 10 LSPs, 5 Areas, Description: This command verifies that ISIS LSP flooding is cleared after re-configuration.
```

Roll-back Procedures

- Step: Restore Original BGP Configuration, Command: restore bgp config, Description: This step restores the original BGP configuration if the fix fails.

- Step: Restore Original OSPF Configuration, Command: restore ospf config, Description: This step restores the original OSPF configuration if the fix fails.

- Step: Restore Original ISIS Configuration, Command: restore isis config, Description: This step restores the original ISIS configuration if the fix fails.